OUTPUT

Output peripherals are connected to a computer and output the results of the computer's processing in various forms, including:

- electronic display
- printed text
- video
- audio
- tactile (touch) forms.

MONITOR

Monitors allow users to see the output from the computer on an electronic display. Features of monitors include:

- screen size, which is measured diagonally
- resolution, which is measured in pixels (see pages 9–10 for information about pixels and screen resolutions)
- energy efficiency measures, such as going into a low-power standby mode if no input is detected.

Some monitors also use touch screen technology (see page pages 26–27 for more information about touch screens).



▲ **Figure 1.61** Monitors have evolved over time to provide better quality and more features

SKILLS COMMUNICATION EXECUTIVE FUNCTION

ACTIVITY

▼ USING MONITORS

1 Discuss the different uses for monitors.

2 Create a table of the uses for monitors with different sizes and resolutions (see *Unit 6 Software skills* (pages 220–223) for more information about creating tables).

PRINTER

There are three types of standard printer: dot matrix, inkjet and laser.



▲ **Figure 1.62** Laser (top left), inkjet (right) and dot matrix (bottom left) printers

GENERAL VOCABULARY

carbon copy a copy of a document made using carbon paper (paper coated in a coloured substance that leaves traces on clean paper)

- **Dot matrix:** These printers are sometimes called 'impact' printers. This is because the print is made by hitting or 'impacting' the paper through a ribbon of ink, and this process is very noisy. The paper often has a number of **carbon copy** layers so it can make several copies of the same document as it prints it.



▲ **Figure 1.63** A dot matrix printer

- **Inkjet printers:** These printers use cartridges containing different coloured ink. Black ink is held in a one cartridge, while the colour cartridge is often split into three colours: cyan, magenta and yellow. Tiny dots of these inks are sprayed onto the paper to form images or text.

DID YOU KNOW?

Many cartridges can be refilled to save money and protect the environment.



▲ **Figure 1.64** Cartridges used in an inkjet printer

SUBJECT VOCABULARY

discharge remove the electrical charge from something
toner a type of ink

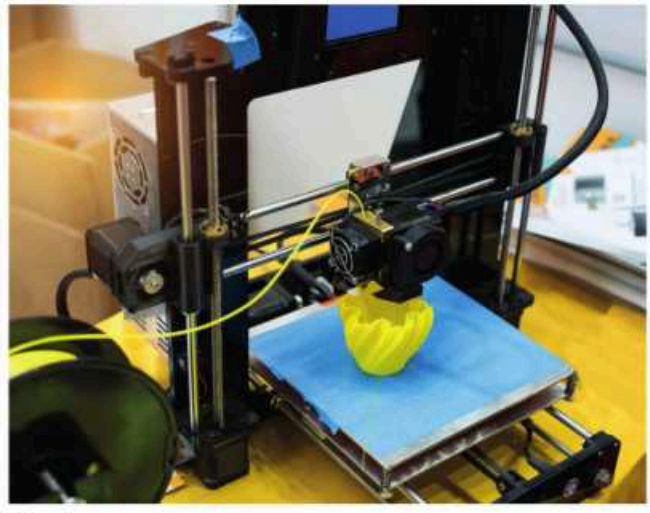
- **Laser printers:** These printers contain a rotating cylinder or drum that holds an electrical charge. A laser is used to **discharge** certain points on the drum and 'draw' an image. Electrically charged **toner** is attracted to those points on the drum and is heated onto paper that is passed over the drum.

▼ Table 1.3 Comparing printer types

	SPEED	PRINTER COST	COST PER COPY	COLOUR PRINTING	QUALITY	EXAMPLES OF WHERE IT IS USED
Dot matrix	Slowest	Cheap	Low	Rare	Low	When multiple copies are required.
Inkjet	Fast	Medium	High	Yes	High	Home printing and photographs
Laser	Fastest	Expensive	Medium	Yes	Highest	High-volume printing, often used in offices

3D PRINTER

Three-dimensional (3D) printers create real-life versions of digital models. These printers work by adding layers of heated material from the bottom of the model all the way to the top. 3D-printed models take a long time to create.



▲ Figure 1.65 A 3D printer

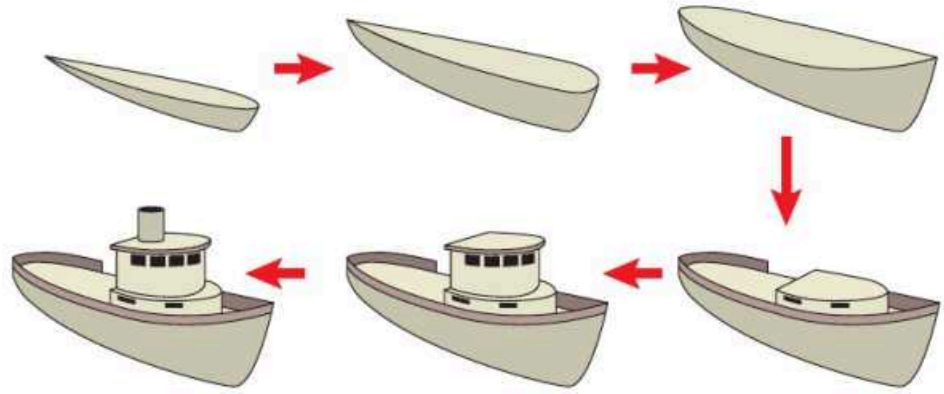
SKILLS INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ USING 3D PRINTERS

Find a video online that shows a 3D printer in action.

3D printers are often used to create complex items, sometimes with working parts. Some of these items could not be assembled in any other way. Examples of 3D-printed items include aeroplane parts, artificial limbs, prototype models of cars and even food such as pasta and chocolate.



▲ **Figure 1.66** Stages of building a 3D-printed model

SKILLS
INTELLECTUAL INTEREST AND CURIOSITY
PERSONAL AND SOCIAL RESPONSIBILITY
ETHICS
COMMUNICATION
REASONING/ARGUMENTATION

ACTIVITY

▼ TECHNOLOGY AND ETHICS

1 In 2013, someone published the plans for a 3D-printed gun online. Discuss the impact of this.

2 Write notes to support two opposing arguments. One argument should support the wide availability of 3D printers, and the other should argue against the wide availability of these printers. You could then use your notes to debate this topic in your class.

PLOTTER

DID YOU KNOW?
Plotters can also use a cutter attachment instead of the pen attachment to cut vinyl. This can be used to create stickers or images to go on printed clothing.



▲ **Figure 1.67** A plotter

GENERAL VOCABULARY
cartographers people who draw and produce maps

Plotters are used by a variety of professionals, including product designers, architects, engineers and **cartographers**. The plotter draws high-quality images created in Computer Aided Design (CAD) software onto large sheets of paper. It does this by moving a pen across the paper to draw the lines, and the pen can be raised away from and lowered onto the sheet of paper. Some plotters can also roll the paper backwards and forwards.

DATA PROJECTOR

Data projectors are used for home entertainment, such as watching television or playing games, and for giving presentations in classrooms and business meetings. They have the following features.

- **Light bulbs:** These bulbs can have various brightness levels (measured in lumens) and are expensive to replace.
- **Resolution:** Like monitors, they use different quality screens (see page 9–10 for more about screen resolutions).
- **Zoom functionality:** The projected image can be made larger by enlarging the image using either **optical zoom** or **digital zoom**.
- **Portability:** Some smaller and lighter projectors are available, and these are easier to move around.

SUBJECT VOCABULARY

optical zoom focusing a lens in order to zoom in on an object
digital zoom enlarging part of an image to produce the effect of zooming in



▲ **Figure 1.68** Data projectors are used to project a large image (or film) onto screens

SKILLS ADAPTABILITY SELF-DIRECTION

ACTIVITY

▼ TYPES OF PROJECTOR

1 Do some online research to find an appropriate projector for:
   - use in a classroom
   - a home entertainment system
   - a person travelling on business.
2 For each projector that you identify, find out how long its light bulbs last (on average) and how much the replacement bulbs cost.

SPEAKER

Speakers allow a computer to output sound. Speakers often come in pairs to provide **stereo sound**. Multiple speakers are commonly used to provide **surround sound** in home entertainment systems. These speakers need a special amplifier that can deliver different levels of sound to each speaker to provide **spatial awareness**.

SUBJECT VOCABULARY

stereo sound sound that comes from two sides (left and right)
surround sound sound that comes from four or more sides, used so that sounds from a film or television programme come from all around the viewer as they would in real life

GENERAL VOCABULARY

spatial awareness the impression of a realistic space



▲ **Figure 1.69** A pair of speakers is required for stereo sound because one is used for the left side and one is used for the right side

SKILLS SELF-DIRECTION
PROBLEM SOLVING

SUBJECT VOCABULARY

mono sound sound that comes from a single speaker

ACTIVITY

▼ DIFFERENT TYPES OF SOUND

Do some research to find out when **mono sound** may be most appropriate. When might listeners prefer stereo sound?

A single speaker is usually used to provide low-quality audio feedback to a user. These speakers are often small and low-quality, and are often found in alert systems such as alarms and buzzers.

SKILLS ADADPTIVE LEARNING

ACTIVITY

▼ USING LOW-QUALITY SPEAKERS

Think of the appliances and devices in your home that use low-quality speakers (you might want to start in the kitchen!).



▲ **Figure 1.70** A piezoelectric 'piezo' speaker used in a watch

CONTROL DEVICE

Control devices are also known as actuators, and they are components of a system that make something happen in the real physical world. These devices are often mechanical. Examples of actuators include:

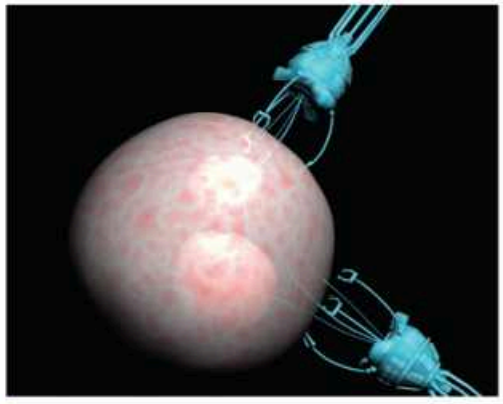
- valves
- pistons
- heaters
- coolers
- motors.



▲ **Figure 1.71** Robots use actuators that are powered in many ways such as electricity, chemicals and compressed air

DID YOU KNOW?

Nanorobots are made up of tiny sensors and actuators that are capable of assisting a surgeon by carrying out very precise medical procedures.



▲ **Figure 1.72** Nanorobots allow doctors to make smaller incisions or cuts and more precise movements than they could if they used their own hands

SECONDARY STORAGE (STORAGE)

GENERAL VOCABULARY

volatile likely to change, especially quickly or unpredictably

HINT

People often confuse storage and memory. You must know the difference between these terms, including what they do and the ways in which they are used.

SUBJECT VOCABULARY

virtual memory space on a computer for storing instructions and programs until they are needed or being used

motherboard the circuit in a computer with connectors to which other components connect

Secondary storage is often referred to as 'storage'. However, it should not be confused with primary storage or main memory, which are terms sometimes used to refer to Random Access Memory (RAM) or Read Only Memory (ROM). You will learn more about RAM and ROM on pages 59–61. Secondary storage is non-**volatile**, which means it will not lose data when the system's power is turned off.

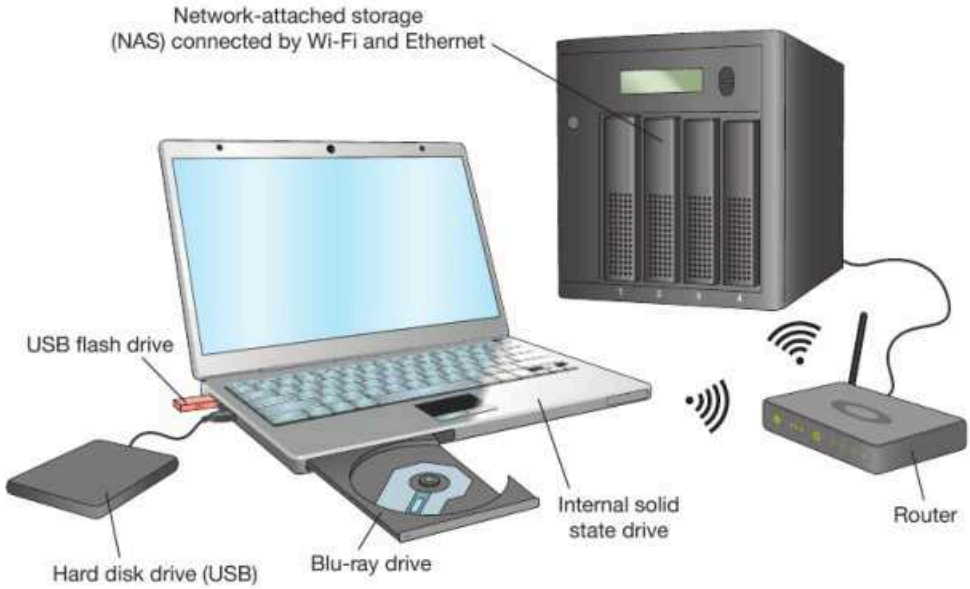
Secondary storage is used for:

- storing a document for future use
- storing an application, ready to be loaded into RAM when the user opens the application
- **virtual memory** (see page 60).

DEVICES

Storage devices are used to store data or software that is used in a computer system.

Storage devices can be either internal or external. Internal drives connect directly to the computer's **motherboard**. External devices are connected to the computer's motherboard through the different ports and adapters on the computer's exterior .



▲ **Figure 1.73** Storage can be internal or external

HINT

You must know the difference between a storage device and storage media. Remember: a storage device *contains* the storage media and *accesses* the data stored on it. For example:

1. flash memory is the media used to store data, whereas a USB drive is used to contain the flash media so that it can be read by the computer after it is plugged into a USB port
2. a DVD disk is the media used to store data, whereas a DVD drive reads the data on the DVD disk.

SUBJECT VOCABULARY

backwards compatibility the ability to be used with older technology without having to be specially adapted

There are different types of storage device.

- Hard disk drives (HDD) contain hard disk media. The drives provide a connection from the disk to the motherboard, either directly or using a wireless adapter like Wi-Fi or a wired port like USB.
- Solid state drives (SSD, often referred to as flash drives) contain flash memory media. They are otherwise identical to hard disk drives.
- Optical disk drives contain optical disk media. Newer drives are often compatible with older media. For example, a disk drive that can read Blu-ray media can also read DVD and CD media. This is known as **backwards compatibility**.

GENERAL VOCABULARY

concentric (of circles or circular objects) sharing the same central point
spindle a rod around which something spins

SUBJECT VOCABULARY

platters circular plates
read/write head the part of a disk drive that passes across (or floats) the platters on a very thin layer of gas above the platters
seek time the time it takes for a read/write head to locate the area on the disk where the data to be read is stored

DID YOU KNOW?

The gap between the platter and the read/write head is only a few nanometers wide. This means that a dust particle or even a smoke particle could cause a collision between the platter and the read/write head. To prevent this, hard drives use filters in their ventilation systems or are built as sealed units.

HINT

If a read/write head crashes into a platter, it can cause damage to data and the platter. This would not happen with a solid state drive.

DID YOU KNOW?

Optical media layers deteriorate (get worse) over time, so they have to be stored carefully to prevent data loss.

DID YOU KNOW?

Blu-ray disks get their name from the colour of laser used. The laser light is actually a violet colour, but perhaps 'violet-ray' was not considered a good enough name.

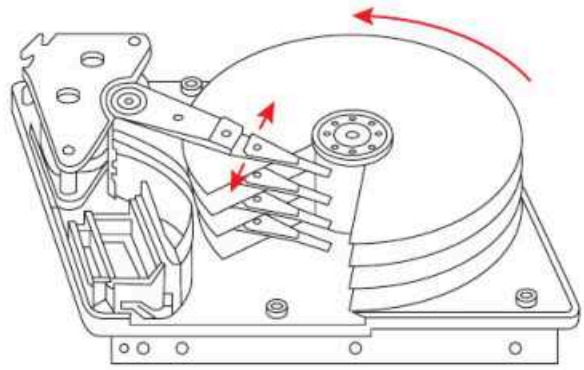
MEDIA

Hard disks

Hard disks are made up of many **concentric platters**. These platters make up a cylinder that spins on a central **spindle**. A **read/write head** moves on an arm across tracks on the platter. The amount of time that it takes the read/write head to access data on the tracks is determined by how fast the cylinder of platters spins and how fast the read/write head is moved across the tracks. When reading, a read/write head changes the magnetic field into electrical current. When writing, it transforms electrical current into a magnetic field.

Typical spin speeds are 5400 revolutions per minute (rpm) or 7200 rpm.

The average **seek time** for a read/write head is 4–15 milliseconds (ms).



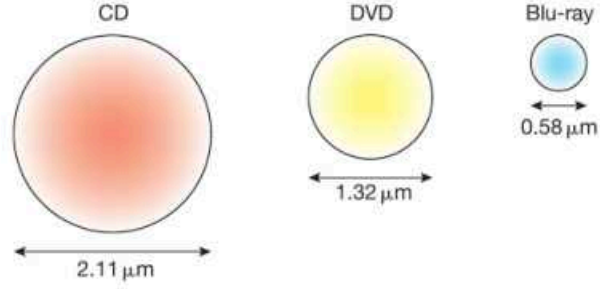
▲ **Figure 1.74** Read/write heads move across a hard disk platter to access data as the cylinder spins

Optical media

Types of optical media include CDs, DVDs and Blu-ray disks.

- **CDs** can store up to 700 MB of data. The data is written to the disk using a laser, which writes data to a plastic layer beneath layers of aluminium and acrylic.
- **DVDs** look very much like CDs. Data is written to DVDs using a shorter wavelength of red laser light, which allows DVDs to store more data. They can store 4.7 GB on a single-sided, single-layer disk. Double-sided, dual-layer disks can store as much as 18 GB.
- **Blu-ray** disks are similar to CDs and DVDs, but use a scratch-protection coating. Violet laser light is used to store data at greater density than the red laser light used in DVDs. They can store 25 GB on a single-sided disk, and double-sided disks can store 50 GB.

All optical media is recordable and is known as CD-R, DVD-R or BD-R. Some CDs and DVDs can be rewritten (CD-RW, DVD-RW), and some Blu-ray media can be written then erased and rewritten (BD-RE).



▲ **Figure 1.75** Comparison of various lasers used to store data on optical media

DID YOU KNOW?

One explanation of why flash memory is so-called is that each cell is flashed (flooded) with electrons, forcing it to hold its charge. One other explanation for the name is that data is written very quickly (or, in colloquial English, 'in a flash').

Flash media

Flash media are more energy-efficient than hard disks as they do not have moving parts. For the same reason, they are also less likely to fail when they are moved around. This makes them suitable for use in portable devices.

Magnetic tape

Magnetic tape was originally designed to record sound, but it is now also used to store data.



▲ **Figure 1.76** Magnetic tape is mainly used for backing up data in large organisations

▼ **Table 1.4** Comparing types of media

	HARD DISK	FLASH MEDIA	OPTICAL MEDIA	MAGNETIC TAPE
DATA ACCESS SPEED	Fast	Fastest	Slow	Slowest
MAXIMUM CAPACITY	Varies, up to 128 GB	Varies, up to many TB	CD 700 MB DVD 18 GB Blu-ray 50 GB	Varies, up to 185 TB
COST PER GB	High	Very high	Medium	Low
USE	Servers, personal computers, backups	Laptops, mobile devices	Multimedia (music, games and films), file backups	Whole system backups and archives
PORTABILITY	Not suitable	Yes	Not suitable	Not suitable

SUBJECT VOCABULARY

data access speed how quickly data can be read from or written to media
capacity the amount of data that can be stored on media
bit (binary digit) a single unit of information with a value of either 0 or 1; eight bits equal one byte

Storage media store data in binary form. This means that each **bit** of data holds one of two values: 0 or 1. Different media types interpret 0 or 1 differently.

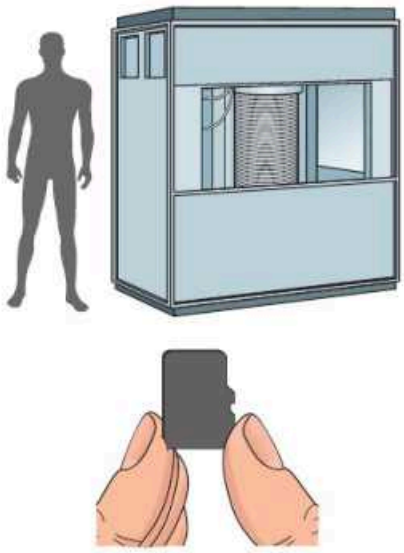
- Hard disks change the magnetic charge of a platter to either negative or positive, depending on whether the value is 0 or 1.
- Optical media use tiny bumps on the disk's surface to represent 0 and 1.
- Flash media use different levels of electrical charge, held in tiny individual cells, to represent 0 or 1.
- Magnetic tape changes the magnetic charge of the tape to either negative or positive, depending on whether the value is 0 or 1.

DID YOU KNOW?

The 'bi' in '**bi**nary' comes from the Latin for 'two'.

HINT

You do not need to understand the units shaded in dark grey in the table for the International GCSE.



▲ **Figure 1.77** The IBM 350 in comparison with a person and a 256 GB microSD card

▼ **Table 1.5** Units of storage capacity

NUMBER OF BYTES	WITH UNIT SYMBOL	NAME
1000	1 KB	Kilobyte
1000,000	1 MB	Megabyte
1000,000,000	1 GB	Gigabyte
1000,000,000,000	1 TB	Terabyte
1000,000,000,000,000	1 PB	Petabyte
1000,000,000,000,000,000	1 EB	Exabyte
1000,000,000,000,000,000,000	1 ZB	Zettabyte
1000,000,000,000,000,000,000,000	1 YB	Yottabyte

The IBM 350 was the first disk drive. It was 172 cm high, 152 cm long and 74 cm wide, and it could store 3.75 MB of data. People could hire one for US$38,400 per year. In 2016, manufacturers started selling the 256 GB microSD card, which is a flash memory card the size of a human fingertip. It was priced at US$175 and can store the same amount of data as 68,266 IBM 350 disks!

DID YOU KNOW?

In 2017, it would take about 9457 trillion years to download one yottabyte of data with a good internet connection.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 State **one** benefit of using barcodes in a supermarket. **(1)**

SKILLS REASONING

2 State **one** reason why small, built-in microphones are used in smartphones and laptops. **(1)**

SKILLS DECISION MAKING

3 Explain why capacitive touch screens do not work when you are wearing woollen gloves. **(2)**

4 State **three** computer systems that **do not** use monitors. **(3)**

SKILLS CRITICAL THINKING

5 State **two** benefits of using laser printers rather than inkjet printers. **(2)**

SKILLS PROBLEM SOLVING

6 State **two** types of business that use plotters. **(2)**

SKILLS REASONING

7 Explain why some 3D printers use biodegradable materials to create models. **(3)**

SKILLS CRITICAL THINKING

8 List **two** sensors used in smartphones. For each one, describe one way in which it is used by the operating system or an app. **(4)**

SKILLS REASONING/ARGUMENTATION

9 Discuss the advantages and disadvantages of using sensors in smart cars. **(8)**

SKILLS DECISION MAKING

10 Explain **one** reason why Blu-ray disks are used to store modern console games. **(2)**